

ST2334 Cheat Sheet, AY26/27 S1

Zhao Wu

September 10, 2026

1 Basic Concepts of Probability

1.1 Experiments, sample spaces, and events

- **Statistical Experiment** is a procedure that produces an observation.
 - **Sample Space** is the set of all possible outcomes of a statistical experiment.
 - **Event** is a subset of the sample space.
 - **Sample Point** is a single outcome of a statistical experiment.
- Sample Point $\subseteq$ Event $\subseteq$ Sample Space

Sure Event and Null Event

- **Sure Event** = Sample Space, S
- **Null Event** = Empty Set / Complement of Sure Event

1.2 Event Operations & Relationships

- **Union** of A and B, $(A \cup B)$
 $A \cup B = \{x \in U \mid x \in A \vee x \in B\}$
- **Intersection** of A and B, $(A \cap B)$
 $A \cap B = \{x \in U \mid x \in A \wedge x \in B\}$
- **Difference** of B minus A, $(B - A)$
 $B - A = \{x \in U \mid x \in B \wedge x \notin A\} = B \cap \bar{A}$
- **Complement** of A ($\bar{A}$)
 $\bar{A} = \{x \in S \mid x \notin A\}$
- **Mutually Exclusive or Disjoint** Events
 $A \cap B = \emptyset$

Contained and Equivalent

- **Contained C**
 $A \subset B \Leftrightarrow \forall x, x \in A \Rightarrow x \in B$
- **Equivalent =**
 $A = B \Leftrightarrow A \subset B \wedge B \subset A$

Subset Relations

No.	Law	Identities
1	Commutative laws	$A \cup B = B \cup A$ $A \cap B = B \cap A$
2	Associative laws	$(A \cup B) \cup C = A \cup (B \cup C)$ $(A \cap B) \cap C = A \cap (B \cap C)$
3	Distributive laws	$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
4	Identity laws	$A \cap U = A$ $A \cap \emptyset = \emptyset$ $A \cup U = U$ $A \cup \emptyset = A$
5	Complement laws	$\overline{\overline{A}} = A$ $A \cap \bar{A} = \emptyset$ $\bar{\bar{U}} = \emptyset$
6	Idempotent laws	$A \cup A = A$ $A \cap A = A$

No.	Law	Identities
7	De Morgan's laws	$\overline{A \cup B} = \bar{A} \cap \bar{B}$ $\overline{A \cap B} = \bar{A} \cup \bar{B}$
8	Absorption laws	$A \cup (A \cap B) = A$ $A \cap (A \cup B) = A$

1.3 Counting Methods

Multiplication and Addition Principles

- **Multiplication Principle**
A sequence of r stages with $n_1, n_2, \dots, n_r$ choices has $n_1 n_2 \dots n_r$ outcomes.
 $n_1 \times n_2 \times \dots \times n_r = n_1 n_2 \dots n_r$
- **Addition Principle**
k non-overlapping procedures with $n_1, n_2, \dots, n_k$ outcomes have $n_1 + n_2 + \dots + n_k$ outcomes in total.
 $n_1 + n_2 + \dots + n_k = n_1 + n_2 + \dots + n_k$

Factorial, Permutations, and Combinations

- **Factorial** ($n!$)
 $n! = n(n-1)(n-2)\dots 2 \cdot 1$
- **Permutation** (P_r^n)
Selection and arrangement of r objects from n objects, where order matters.
 $P_r^n = \text{frac}\{n!\}\{(n-r)!\}$
- **Combination** ($\binom{n}{r}$)
Selection of r objects from n objects, where order does not matter.
 $\binom{n}{r} = \text{frac}\{n!\}\{r!(n-r)!\}$

1.4 Probability

Probability of an event A is a measure of the likelihood of the event occurring.

Relative Frequency

Suppose an experiment is repeated n times and event A occurs m times. The relative frequency of A is given by:

$$P(A) = \lim_{\{n \rightarrow \infty\}} \frac{m}{n}$$

As n increases, the relative frequency of A approaches the probability of A.

Axiomatic Definition of Probability

A probability function P is a function that assigns a number P(A) to each event A in the sample space S, such that:

$$0 \leq P(A) \leq 1$$

$$P(S) = 1$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Properties of Probability

Prop.	Name	Statement
1.4	Empty set	$P(\emptyset) = 0$
1.5	Finite additivity	If $A_i \cap A_j = \emptyset$ for $i \neq j$, then $P(A_1 \cup \dots \cup A_n) = P(A_1) + \dots + P(A_n)$
1.6	Complement	$P(\bar{A}) = 1 - P(A)$
1.7	Partition of A	$P(A) = P(A \cap B) + P(A \cap \bar{B})$
1.8	Inclusion-exclusion	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
1.9	Monotonicity	$A \subset B \Rightarrow P(A) \leq P(B)$

Prop.	Name	Statement
1.10	Union bound (Boole's inequality)	$P(A_1 \cup \dots \cup A_n) \leq P(A_1) + \dots + P(A_n)$

1.5 Conditional Probability

- **Conditional Probability** ($P(A \mid B)$)
Probability of event A occurring given that event B has occurred.

Definition of Conditional Probability

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

Simpson's Paradox

A phenomenon where a trend appears in different groups of data but disappears or reverses when the groups are combined. Or formally,

$$P(A \mid B \cap C_i) \geq P(A \mid B' \cap C_i) \forall i \text{ but } P(A \mid B) < P(A \mid B')$$

Multiplication Rule

$$P(A \cap B) = P(A \mid B)P(B) \text{ if } P(B) > 0$$
$$\text{or } P(A \cap B) = P(B \mid A)P(A) \text{ if } P(A) > 0$$

Inverse Probability Formula

$$P(A|B) = \frac{P(A)P(B|A)}{P(B)}$$

1.6 Independence

Events A and B are independent, $A \perp B \Leftrightarrow$

$$P(A \cap B) = P(A)P(B)$$

Tautologies:

- Suppose $P(A) > 0$ and $P(B) > 0$. If $A \perp B$, then A and B are not mutually exclusive (aka $A \cap B \neq \emptyset$)
- Suppose $P(A) > 0$ and $P(B) > 0$. If A and B are mutually exclusive, then $A \not\perp B$.
- $S \perp A$ for all events A
- $\emptyset \perp A$ for all events A
- If $A \perp B$, then $A \perp B'$, $A' \perp B$, $A' \perp B'$
- Independence cannot be expressed in terms of venn diagram

1.7 Law of Total Probability

Suppose $A_1, A_2, \dots, A_n$ is a partition of the sample space S

$$P(B) = \sum_{i=1}^n P(B \cap A_i) = \sum_{i=1}^n P(B \mid A_i)P(A_i)$$
$$P(B) = P(A)P(B \mid A) + P(A')P(B \mid A')$$

1.8 Bayes' Theorem

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{P(B \mid A)P(A)}{\sum_{i=1}^n P(B \mid A_i)P(A_i)}$$

2 Random Variable

Random Variable is a function that transform a sample from sample space to a real number.

$$X: S \rightarrow \mathbb{R}: s \in S \text{ and } X(s) = x \in \mathbb{R}$$

Range space of X is set of real numbers that satisfies

$$R_X = \{x \mid X(s) = x \forall s \in S\}: R_X \subseteq \mathbb{R}$$

Subsets of Sample Space

Set of all sample such that the result of its random variable is x.

$$\{X = x\} = \{s \in S : X(s) = x\}, \{X = x\} \in S$$

Set of all sample such that the result of its random variable is in A.

$$\{X \in A\} = \{s \in S : X(s) \in A\}, \{X \in A\} \in S$$

2.1 Probability Distribution

Probability Function (pf) or Probability Mass Function (pmf)

Probability Function (pf), f(x) is defined as probability for $\{X = x\}$ if $x \in R_X$

$$f(x) = \begin{cases} P(X = x) & \mid \forall x \in R_X \\ 0 & \mid \forall x \notin R_X \end{cases}$$

Discrete Random Variable

Discrete random variable, the number of R_X is finite or countable. $R_X = \{x_1, x_2, \dots\}$

A well defined pf for a discrete random variable X satisfies all the following condition:

1. $f(x_i) \geq 0, \forall x_i \in R_X$
2. $f(x_i) = 0, \forall x_i \notin R_X$
3. $\sum_{i=1}^{\infty} f(x_i) = 1$ or $\sum_{x_i \in R_X} f(x_i) = 1$

Let $B \in R_X$

$$P(X \in B) = \sum_{x_i \in B \cap R_X} f(x_i)$$

Continuous Random Variable

Continuous random variable, R_X is an interval or a collection of intervals

A well defined pf for a continuous random variable X satisfies all the following condition:

1. $f(x) \geq 0 \forall x \in R_X$ and $f(x) = 0 \forall x \notin R_X$
2. $\int_{R_X} f(x) dx = \int_{-\infty}^{\infty} f(x) dx = 1$

For any a and b such that $a \leq b$

$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

2.2 Cumulative Distribution Function

Cumulative Distribution Function, F(x) for any random variable X is defined as

$$F(x) = P(X \leq x) = \begin{cases} f(0) + f(1) + \dots + f(x) & \text{if X is discrete} \\ \int_{-\infty}^x f(t) dt & \text{if X is continuous} \end{cases}$$

Let $a < b$

$$P(a \leq X \leq b) = P(X \leq b) - P(X < a) = F(b) - F(a -)$$

$$F(a -) = \lim_{x \rightarrow a+} F(x)$$

$$f(x) = F'(x) - F'(x -)$$

$$F(x) = \int_{-\infty}^x f(t) dt$$

$$f(x) = \frac{d}{dx} F(x)$$

$$F(b) - F(a) = P(a < x < b)$$

2.3 Expectation and Variance

Expectations

Expectations or mean of X is defined by

$$\mu_X = E(X) = \sum_{x_i \in R_X} x_i f(x_i)$$
$$= \int_{x \in R_X} x f(x) dx$$

μ_X may not be in R_X

Properties of Expectation

1. $E(aX + b) = aE(X) + b$
2. $E(X + Y) = E(X) + E(Y)$
3. $E[g(X)] = \sum_{x \in R_X} g(x)f(x) = \int_{R_X} g(x)f(x)dx$

Variance

Variance of X is defined as μ_X^2

$$\begin{aligned}\mu_X^2 &= V(X) = E[(X - \mu_X)^2] \\ &= \sum_{x \in R_X} (x - \mu_X)^2 f(x) \\ &= \int_{-\infty}^{\infty} (x - \mu_X)^2 f(x)dx \\ &= V(X) = E(X^2) - [E(X)]^2 \\ \text{where } E(X^2) &= \sum_{x_i \in R_X} x_i^2 f(x_i)\end{aligned}$$

Alternatively,

Standard deviation of X is defined as σ_X

$$\sigma_X = \sqrt{V(X)}$$

Properties of Variance

- $V(aX + b) = a^2 V(X)$